

This exam is closed books. Write your name on every page. Write clearly and legibly. Explain your work in words.

P1 (0.5p). A system with the input signal $r(t)$ and the output $y(t)$ is described by the differential equation:

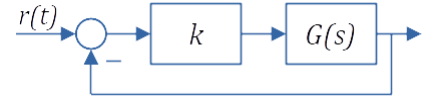
$$\frac{d^3 y(t)}{dt^3} + \frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - y(t) = \frac{du(t)}{dt}$$

a) (0.3p) Determine the transfer function of the system.

b) (0.2p) Analyze the stability of the system and justify your answer.

P2 (1.5p). For the closed-loop system shown in the figure, where:

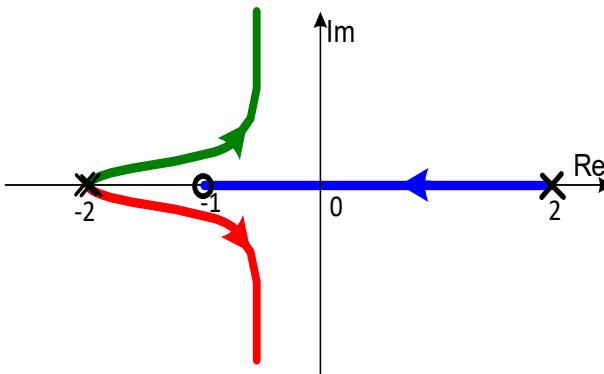
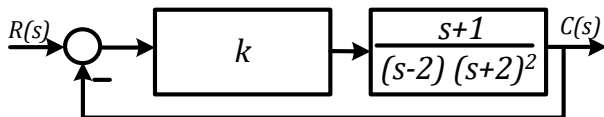
$$k > 0, \quad G(s) = \frac{1}{8s + 1}$$



a) (1p) Determine the value of the parameter k so that the settling time of the closed-loop system is $t_s = 4$ seconds. How will the settling time change if the value of k is increased beyond the one calculated previously? Sketch two step responses with different settling times for this system and mark the settling time.

b) (0.5p) Choose a value for k so that the closed-loop system is stable and find the steady-state error for a step input $r(t) = 4, t \geq 0$.

P3 (2p). A negative feedback closed-loop system has the block diagram and the root locus given below.



a) (1p) Analyze the stability of the closed-loop system using the root locus:

- Determine the range of values for the parameter k that ensure the closed-loop system is stable (*use the root locus*).
- Explain how the location of the closed-loop poles varies with k and how this affects the stability of the system.

b) (0.5p) Determine the range of values (if any) for k so that the closed-loop system is underdamped or overdamped, and explain your conclusion based on the root locus.

c) (0.5p) Choose a value for k that ensures the closed-loop system is stable and calculate the steady-state value of the output when the input is $r(t) = 1$.

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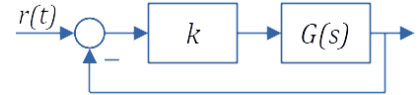
$$\frac{d^3 y(t)}{dt^3} + \frac{d^2 y(t)}{dt^2} + \frac{dy(t)}{dt} + y(t) = \frac{du(t)}{dt} + u(t)$$

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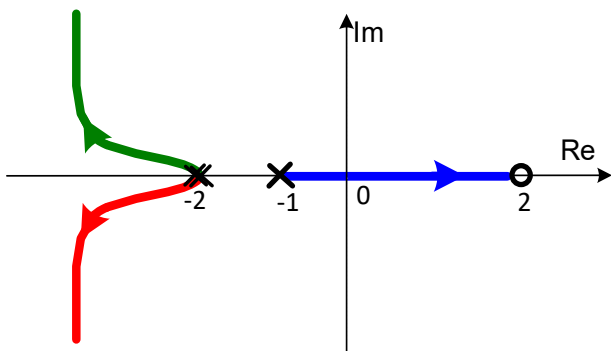
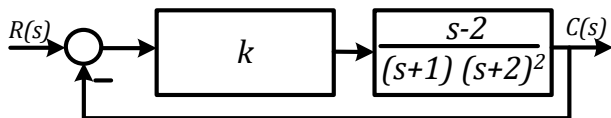
$$k > 0, \quad G(s) = \frac{1}{s+3}$$



a) (1p) Determine the value of the parameter k so that the settling time of the closed-loop system is $t_s = 0.5$ seconds. How will the settling time change if the value of k is increased beyond the one calculated previously? Sketch two step responses with different settling times for this system and mark the settling time.

b) (0.5p) Choose a value for k so that the closed-loop system is stable and find the steady-state error for a step input $r(t) = 4, t \geq 0$.

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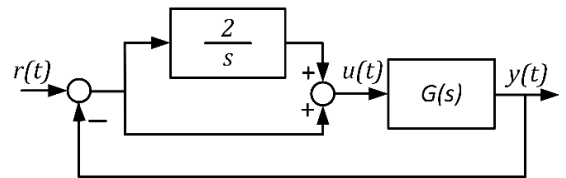
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c) (0.5p) Choose a value for k that ensures the closed-loop system is stable and calculate the steady-state value of the output when the input is $r(t) = 1$.

P1 (1.5p). Consider the closed-loop system shown in the figure:

a) (0.5p) Determine the transfer function $G(s)$ if the differential equation that describes the relationship between the input $u(t)$ and the output $y(t)$ of this block is:

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 2y(t) = \frac{du(t)}{dt}$$



b) (0.5p) Compute the equivalent closed-loop transfer function from the input $r(t)$ to the output $y(t)$.

c) (0.5p) Determine the steady-state error for a unit step input $r(t) = 1, t \geq 0$.

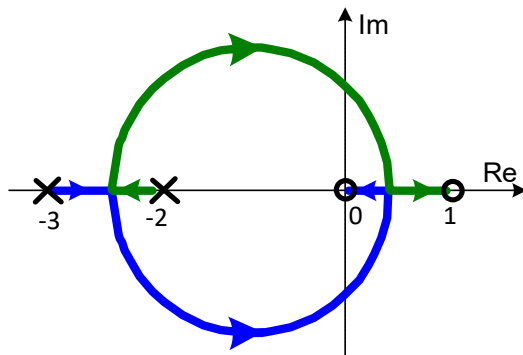
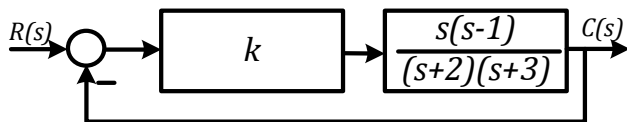
P2 (1p). Sketch the step response for the following systems with no zeros and the poles:

(Sketch only the axes and the general shape of the step response)

$G_1(s): p_1 = 1, \quad p_2 = -1$	$G_2(s): p_1 = 2j, \quad p_2 = -2j$	$G_3(s): p_1 = -1, \quad p_2 = -2$
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For each system, determine whether it is stable, critically stable, or unstable, and classify it as underdamped, overdamped, or undamped.

P3 (1.5p). A negative feedback closed-loop system has the block diagram and the root locus given below.

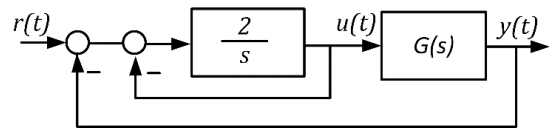


a) (1p) Analyze the stability of the closed-loop system using the root locus:

- Determine the range of values for the parameter k that ensure the closed-loop system is stable (use the Routh-Hurwitz criterion)
- Explain how the location of the closed-loop poles varies with k and how this affects the stability of the system.

b) (0.5p) Choose a value for k that ensures the closed-loop system is stable and calculate the steady-state value of the output when the input is $r(t) = 1$.

P1 (1.5p). Consider the closed-loop system shown in the figure:



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b) (0.5p) Compute the equivalent closed-loop transfer function from the input $r(t)$ to the output $y(t)$.

c) (0.5p) Determine the steady-state error for a unit step input $r(t) = 1, t \geq 0$.

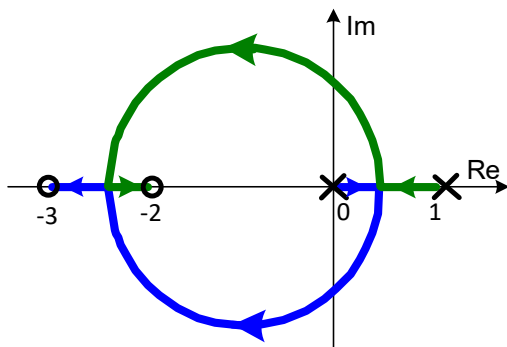
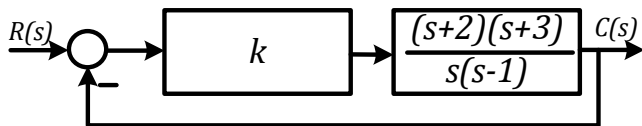
P2 (1p). Sketch the step response for the following systems with no zeros and the poles:

(Sketch only the axes and the general shape of the step response)

$G_1(s): p_1 = -3, \quad p_2 = -1$	$G_2(s): p_1 = -1 + 2j, \quad p_2 = -1 - 2j$	$G_3(s): p_1 = 2j, \quad p_2 = -2j$
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For each system, determine whether it is stable, critically stable, or unstable, and classify it as underdamped, overdamped, or undamped.

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- Determine the range of values for the parameter k that ensure the closed-loop system is stable (use the Routh-Hurwitz criterion)
- Explain how the location of the closed-loop poles varies with k and how this affects the stability of the system.

b) (0.5p) Choose a value for k that ensures the closed-loop system is stable and calculate the steady-state value of the output when the input is $r(t) = 1$.